

Sorting, Price Informativeness, and Information Acquisition in Dark Markets ^{*}

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Abstract

We design an experiment to study the effects of dark trading on incentives to acquire costly information, price efficiency, market liquidity, and investors' earnings in a financial market. When the information precision is high, adding a dark pool alongside a lit exchange encourages information acquisition but negatively impacts liquidity at the lit market. In line with theory, investors with stronger signals use the lit exchange relatively more. Dark trading is associated with improving price quality as long as the dark pool activity remains below a threshold. Uninformed traders are better off when they trade more in the dark pool.

Keywords: Market institutions, dark pools, information aggregation, the efficiency of security markets, costly information acquisition, experiments

JEL Classification Numbers: C91, C92, G12, G14

In the last few decades, there has been a proliferation of equity trading systems, among which dark pools have rapidly grown in popularity.¹ In a dark pool, investors can buy and sell stocks without publicly displaying their orders. As opposed to traditional stock exchanges, these dark venues lack pre-trade transparency. Traders in dark pools submit buy and (or) sell orders, and trades are executed using prices derived from the exchanges. As markets are becoming fragmented and a significant part of liquidity is increasingly unobserved, it questions the long-held view that prices are a valuable source of information (Hayek (1945), Fama (1970)). Would prices at the exchanges continue to convey meaningful information? Indeed, dark pools have raised regulatory concerns in that they may harm price discovery in financial markets.²

While dark trading has existed in stock markets for quite some time now in the form of either over-the-counter (OTC) decentralized trading (Duffie, Gârleanu, and Pedersen (2005), Duffie, Malamud, and Manso (2009), Duffie (2012), Duffie, Malamud, and Manso (2014), Malamud and Rostek (2017), Rostek and Yoon (2020)) or special hidden order types on exchanges (Blume and Goldstein (1997), Madhavan, Porter, and Weaver (2005), Baruch (2005), Boulatov and George (2013), Buti and Rindi (2013)), the emergence of dark pools as alternative trading system operating fully outside transparency requirements has attracted enormous trade volumes. Sev-

¹Dark pools accounted for 13.66% of US equity trading volume in Nov. 2021, as opposed to 7.51% in 2008 (Rosenblatt Securities: Let There Be Light - US Edition). In Nov. 2021, dark venues executed 18.84% of pan-European on-venue turnover, and dark MTFs hit a market share high (Rosenblatt Securities: Let There Be Light - European Edition).

²For example, in 2014, then-SEC-Chair Mary Jo White remarked that “we must continue to examine whether dark pool trading volume is approaching a level that risks seriously undermining the quality of price discovery provided by lit venues.” Before the Securities Traders Association’s 82nd Annual Market Structure Conference in Sep. 2015, then-SEC-Commissioner Kara M. Stein mentioned that “as more and more trading is routed to dark venues that have restricted access and limited reporting, I am concerned that overall market price discovery may be distorted rather than enhanced.” As recent as Aug. 2021, on CNBC’s Squawk Box, the SEC-Chair Gary Gensler pointed out that dark pools have been increasingly common during the recent rise in retail investing and asserted that the agency is “looking very closely” at market structure and dark pools of trading hidden from the public until execution to promote transparency. He was responding to allegations from retail traders that hedge funds and other “big actors” may be manipulating markets.

eral of these dark pools are broker-dealer-owned and are typically run by investment banks (CrossFinder, Sigma X, Citi-Match, MS Pool). In contrast, others are operated by independent companies (Instinet, Liquidnet, ITG Posit).

Given the prominence of the issue, there has been significant interest within academic research in recent years to investigate the effect of dark trading on the informational efficiency of prices. Recent theoretical studies have argued that dark markets can improve price efficiency in the lit exchange due to a sorting effect such that those with the strongest private information sort into the lit market because trading in the dark market may result in trades not being executed (Zhu (2014), Ye (2016)). This non-execution risk arises because traders with strong signals would want to trade in the same direction, resulting in fewer trading opportunities if they trade in the dark market where there is no market maker. Whether or not investors respond to the non-execution risk and sort according to the theory is ultimately an empirical question.

In this paper, we examine the effect of introducing a dark pool alongside a lit exchange on investors' decisions to invest in information production and the choice of venue by informed and uninformed traders. We also investigate how the presence of a dark venue for trading affects overall market performance as captured by various informational efficiency and liquidity measures. Specifically, we ask the following questions. Given that markets are now fragmented, and a substantial portion of the liquidity is anticipated to be hidden, does the incentive to acquire costly information about stock fundamentals increase compared to the benchmark of a centralized trading institution with full pre-trade transparency? How does the relative usage of the dark pool vis-à-vis the lit exchange depend on the relative strength of private information held by an investor? Are informational efficiency of prices and market quality necessarily degraded when dark trading occurs?

We design an experimental asset market with the endogenous acquisition of costly information to address the above questions. We assume two equally likely states of nature, A and B , and a single asset, namely, an Arrow-Debreu security that provides a payoff only in state A . Before trading, some individuals can acquire costly, imperfect signals about the state of nature. Signals are binary and are independent and identically distributed (i.i.d.), conditional on the state. In the experiments, we implement markets with either low or high precision of signals. We investigate two market structures, one with a single lit exchange where all order submissions by a trader are observable to other traders and another where two parallel trading venues exist. In the latter institution, in addition to the lit exchange, traders can submit orders and transact in a dark pool where order submissions of other investors are unobservable. Transaction prices in the dark pool are derived from the existing buy and sell offer prices in the lit exchange.

Experiments provide a useful complement to theoretical and empirical studies on the consequences of dark pool trading. In the laboratory, one can employ a trading mechanism close to the one used in actual markets while still having the ability to control and change variables to allow clean causal inferences. The novelty of our analysis stems from the fact that we consider fragmented markets in the laboratory where traders can exchange an Arrow-Debreu security either in a lit market with a fully transparent limit order book or in a dark pool where order submissions are hidden.

The theoretical prediction regarding the self-selection of venues and sorting of traders based on their private information finds support in our data when the information precision of signals is high. Informed traders want to trade in the same direction and suffer lower execution probabilities in the dark pool. As a result, they are more likely to migrate to the lit exchange for faster execution.

Our experiments further show that the effect of dark trading on the informational efficiency of prices and market outcomes depends on the information precision of signals. While a significant portion of liquidity on the lit exchange is substituted away to the dark pool (the *crowding out effect*) irrespective of the information precision of signals, adding a dark pool has two positive effects when the information precision is sufficiently high. First, investors acquire more signals in the presence of a dark pool (*information acquisition effect*). Second, traders with stronger signals use the lit exchange relatively more than those with moderate or uninformed signals (*sorting effect*).

We find little evidence that adding a dark pool alongside a lit exchange alters the quality of prices, on average. While the addition of a dark pool does not affect the aggregate trading volume in the market, market depth at the exchange declines. Additionally, quoted and effective bid-ask spreads increase with dark trading only when information precision is high. Therefore, the overall effect of dark trading on aggregate outcomes is significant when the signal informativeness is high and is somewhat muted for markets with low precision of signals.

With high precision of signals, at a modest level of dark trading, the combination of *information acquisition effect* and *sorting effect* outweighs the *crowding out effect*. As the dark market participation goes up, the *crowding out effect* grows stronger and eventually exceeds the positive effects. Consequently, compared to the benchmark of trading in a single lit exchange, the ability of prices to aggregate existing information and forecast the underlying state of nature first improves and then deteriorates as the dark trading volume as a percentage of total volume increases. This non-linear association suggests that the price-quality improves against the baseline of no dark trading at a modest level of dark trading. Therefore, our results strongly support the view that there is a concern with dark trading affecting price discovery only if it

exceeds a certain threshold.

We also find that the addition of a dark pool in itself does not affect the earnings of either the uninformed or informed traders. In general, with a single lit exchange or with two parallel trading venues, those with strong signals are able to outperform those who are uninformed. Information rents are greater when information precision is low as prices at the lit exchange reveal less information regarding the underlying state of nature. Furthermore, in the presence of a dark pool, there is a positive relationship between the relative usage of the dark pool and earnings for uninformed traders. Still, no such association exists for informed traders. This points toward an adverse selection risk for the uninformed investors trading primarily at the lit exchange and indicates that adding a dark venue allows some uninformed traders to mitigate part of the risk.

Our research complements empirical papers that have yielded conflicting results on the impact of dark pools on price and market quality measures. For example, while Buti, Rindi, and Werner (2011), Jiang, McInish, and Upson (2012), Fleming and Nguyen (2013) find that dark trading improves price discovery, several studies find that dark trading has a negative effect on price discovery (Hendershott and Jones (2005)) and harms overall market quality (Hatheway, Kwan, and Zhen (2017)). Albuquerque, Song, and Yao (2020) demonstrate that stocks subjected to the “trade at” provision experience larger price errors, suggesting that dark trading improves informational efficiency on an intraday basis. Comerton-Forde and Putniņš (2015) report that low levels of non-block dark trading are benign or even beneficial for informational efficiency, but high levels are harmful. Foley and Putniņš (2016) find that dark limit order markets benefit market quality and informational efficiency, but dark midpoint crossing systems do not significantly affect market quality.

Relatedly, various empirical papers find evidence that informed traders utilize

dark pools (Boulatov and George (2013), Reed, Samadi, and Sokobin (2020), Ye and Zhu (2020), Hu, Jones, and Zhang (2021)), and this may have implications for price discovery at the exchanges. Finally, a recent study by Brogaard and Pan (2022) reports that a higher level of dark pool trading is associated with more information acquisition.³ Our experimental work analyzes the causal relationship between dark trading and costly information acquisition and complements their study.

In laboratory markets, hidden liquidity has been investigated in a single limit order book environment by providing traders with the ability to hide orders (Bloomfield, O’Hara, and Saar (2015), Gozluklu (2016)). Bloomfield, O’Hara, and Saar (2015) report that order strategies are greatly affected by allowing hidden liquidity, with traders substituting nondisplayed for displayed shares and changing the aggressiveness of their trading. Additionally, most aggregate market outcomes, such as liquidity and informational efficiency, are not affected. Using iceberg markets, Gozluklu (2016) finds that, without information friction, market opacity enhances liquidity. Under informed trading, adverse selection drives market outcomes mainly around news announcements.⁴ In contrast to these studies, we implement fragmented markets with separate lit and dark venues and allow for endogenous information acquisition.

Finally, another recent paper by Halim et al. (2025) also implements asset markets having parallel venues for trading. However, our study focuses on the choice of venue by traders based on their relative strength of private information and how the addition of a dark venue affects information acquisition. In contrast, Halim et al. (2025) are primarily interested in understanding the effects of dark trading on allocative efficiency as a function of the distribution of fundamental information among traders

³Brogaard and Pan (2022) utilize weekly dark pool trading data from September 2014 to December 2017 provided by the Financial Industry Regulatory Authority (FINRA) and use Weller’s (2018) jump ratio and the future earnings response coefficient (FERC) as two measures of information acquisition.

⁴Iceberg markets allow both displayed and partially displayed orders (i.e., a minimum size must be displayed, and the remainder can be nondisplayed).

differing in dividend returns.

The literature on costly information acquisition in laboratory markets has shown that the market value of information approaches zero when traders submit sealed bids in an environment with perfect information (Copeland and Friedman (1992), Sunder (1992)). In contrast, in a setting where private information is imperfect, it is valued by the market participants (Ackert, Church, and Shehata (1997)). Huber, Angerer, and Kirchler (2011) demonstrate that it is possible for informed traders to obtain lower net profits on average compared to uninformed traders.

Page and Siemroth (2017) show that traders are more likely to acquire costly information if they have a larger endowment in terms of cash and assets if their existing information is inconclusive, and if they are less risk-averse. Halim, Riyanto, and Roy (2019) demonstrate that social communication among investors results in lower information acquisition due to a trader's temptation to free-ride on the signals purchased by her neighbors. A recent study by Asparouhova, Bossaerts, and Yang (2024) reports that participants with no initial information in decentralized markets remain willing to pay for information, in contrast to the centralized markets in Sunder (1992). To the best of our knowledge, ours is the first paper to study the question of how the presence of a dark pool affects price efficiency and market outcomes with costly and endogenous information acquisition in experimental asset markets.

The remainder of the paper is organized as follows. Section I provides the theoretical background motivating our experimental design. Section II describes our experimental markets, the treatments, and the controls that constitute our experimental design. The results are presented in Section III. In Section IV, we provide a discussion relating our experimental findings to the theoretical predictions and summarize the mechanism underlying the results. The last section concludes.

I. Background

Our experiments are motivated by the theoretical models that study traders’ venue choice and price discovery when a dark pool is added alongside a lit exchange. Zhu (2014) and Ye (2016) are the closest theoretical papers.⁵ Using a model of strategic venue selection by informed and liquidity traders, Zhu (2014) shows that relatively more informed traders are pushed into the lit exchange while relatively more uninformed traders go to the dark pool. As a result of this self-selection, there is a reduction in the noisiness of demand and supply on the exchange, which improves price discovery.

The self-selection result in Zhu (2014) occurs because of the difference in execution risk of informed orders and liquidity orders in the dark pool. Because informed orders are positively correlated with the asset’s value and, therefore, with each other, informed traders would want to trade in the same direction and suffer lower execution probabilities in the dark pool. On the other hand, liquidity orders are less correlated and have higher execution probabilities in the dark pool.

While Zhu (2014) considers perfectly informed traders, Ye (2016) assumes that informed traders receive noisy and heterogeneous signals about the asset’s fundamental value and characterizes an equilibrium where traders with strong signals trade in exchanges, traders with moderate signals trade in dark pools, and traders with weak signals do not trade. As a result, when information precision is high, most informed traders trade in the exchange, and adding a dark pool enhances price discovery. On the other hand, when information precision is low, the majority of the informed traders trade in the dark pool, and adding a dark pool impairs price discovery.

⁵Another related paper is by Ye (2011), which studies the venue choice of a large informed trader in the Kyle (1985) framework. Ye (2011) assumes that only the informed trader can freely select trading venues and finds that the addition of a dark pool harms price discovery on the exchange. In contrast, we allow both informed and uninformed traders to select venues in our experiments, following Zhu (2014) and Ye (2016).

The theory, therefore, clearly identifies the precision of information signals as an important determinant of how dark pools affect price discovery and puts forth an underlying mechanism based on the dark pool participation as a function of the strength of the private information held by an investor. Although, given the complexity of the theoretical model, we do not implement the model *ad verbatim*, the basic environment of the theory is retained such that the central hypotheses can be tested using our experimental data. As we describe in the next section, trading in our study was organized as electronic limit order markets, traders in our experiment ended up with noisy and heterogeneous signals about the fundamental value, and we implemented treatments with low and high precision of such signals.

II. Experimental Design

A. General Structure

The data for this study were gathered from 24 experimental sessions conducted at Nanyang Technological University (NTU), Singapore. We had 288 participants in total, with 12 participants in each session. Participants were recruited from the population of undergraduate and graduate students at NTU from various majors ranging from Social Sciences, Business and Economics, Humanities, Engineering, and Sciences. No subject participated in more than one session of this experiment. Sessions lasted approximately two hours, and participants earned on average S\$19.02 in addition to a show-up fee of S\$2.⁶

Upon arrival, subjects were seated at visually isolated computer workstations. Instructions were read aloud, and subjects also received a copy of the instructions.⁷ Participants were prohibited from talking during the experiment, and all commu-

⁶Payoffs, inclusive of the show-up fee, ranged from S\$9 to S\$30.

⁷A sample copy of the instructions is provided in the Appendix.

nication took place via the experimental software. Each session consisted of three practice periods and 20 main periods.⁸ Activity during the practice periods did not count toward final earnings.

We employed the ball-and-urn setup of the experiments conducted by Anderson and Holt (1997).⁹ At the start of each period, a virtual urn (A or B) was randomly selected by the computer, with each urn having an equal chance of being chosen. Both types of urn contained ten balls in total, with each ball being of either black or white color. Urn A contained more black balls than urn B, while urn B contained more white balls than urn A. All of this information was common knowledge to the participants. The realization of the urn was fully revealed to the subjects only at the end of a period.

Traders had the opportunity to exchange several units of a financial asset every period by participating in a virtual financial market. All accounting and trading were done in experimental currency units (ECU). The market was implemented using the z-Tree computer program (Fischbacher (2007)). At the end of each period, one unit of the asset paid a dividend of either 10 ECU if the underlying urn was A or 0 ECU if the urn was B.

In each period, all participants started with the same initial endowment of 600 ECU and 40 assets. The endowment and earnings from one period could not be carried forward to the next period; that is, each period was independent of the other. Prior to trading, some participants could acquire private information at a cost. This information was provided in the form of balls drawn independently and with replacement from the underlying urn; that is, each signal was i.i.d., conditional on the underlying

⁸At the end of the instructions phase and prior to the start of the experiment, all participants had to complete a quiz to ensure that they understood the concepts and instructions required for the experiment. We started the experiment only after everyone in the room answered all quiz questions correctly.

⁹We followed Page and Siemroth (2017) and Halim, Riyanto, and Roy (2019) in explaining the ball-and-urn setting to the participants.

state of nature. Out of the twelve traders in a market, only eight were allowed to acquire costly information. These potentially informed traders were given 60 seconds during which they could acquire up to ten draws at the cost of 6 ECU each. The remaining four traders were uninformed, which implies that the only information they possess is the prior probability that each of the urns has an equal likelihood of being selected as the underlying urn.¹⁰

The ball draws revealed to participants provided them with information about the underlying state of nature and, hence, about the value of the asset. For instance, observing more black ball draws would indicate that urn A was more likely to be the underlying urn. Each informed trader has imperfect information about the asset's value, and thus, trading on private information is still risky for informed traders. In the instructions, we briefly explained the concept of posterior probability and the procedure for computing the posterior. Participants were not required to calculate the posterior themselves. Instead, the computer program displayed the posterior for each subject according to their ball draws. For the uninformed traders, traders who could acquire information but decided not to, and traders who drew an even number of balls with the same number of balls of each color, the posterior remained at the prior level of 0.5.

After the information-acquisition stage was over, participants entered the trading stage. A trading phase lasted for two minutes, within which all subjects were free to purchase and sell units of the asset at any time, provided that they did not violate the short-selling (negative holdings) constraint. In addition, subjects were required to maintain a positive cash balance to make any purchases. If engaging in a trade would violate either the short-sale or cash-balance constraint, the computer program prohibited individuals from doing so. Throughout the trading stage, pertinent infor-

¹⁰Subjects knew their type: uninformed or (potentially) informed, as well as the fact that there were a total of four uninformed traders and eight potentially informed traders in a market. They also knew that everyone had the same initial endowment of cash and assets.

mation such as the profile of draws revealed to them and the posterior probability of the underlying urn being A given their draw profiles (only for the informed traders), as well as their ECU and asset balance available for trading were displayed on a participant’s trading window.¹¹ Once trading closed, the underlying urn was revealed together with the subject’s earnings and the average transaction price in the period.

Following completion of the last period, subjects were required to participate in a standard risk-elicitation task (Holt and Laury (2002)). Participants were also asked to answer a questionnaire aimed at collecting additional information such as gender, age, prior trading experience, study background, etc. At the end of the experiment, the program randomly selected three of the 20 periods for the purpose of payment. Subjects were paid the average of the payouts from these three periods.

B. Treatments

We implemented four treatments with a 2×2 between-subject design. We varied the trading institution by having only a lit market or adding a dark pool alongside a lit exchange and the information precision of the privately acquired signal. This choice of treatments allows us to study the implications of dark trading for sorting behavior, market quality, and information acquisition, conditional on the precision of informative signals available in the market. The experimental design is summarized in Table I.

Under the *Lit Only* trading institution, the market was organized as a multiple-unit double auction (MUDA) market. It was set up like a typical electronic limit order book where traders can enter buy or sell limit orders. Limit orders to buy or sell a security had prices between 0 and 10 ECU.¹² All buy/sell offers were publicly displayed on the order book. Once a trader entered an order, the book of publicly

¹¹For the uninformed traders, the computer screen mentioned that the likelihood of urn A is 50%, equaling the prior.

¹²Subjects could place limit orders with offer prices rounded up to one decimal place.

Table I
Experimental Design: Summary of Treatments

This table presents our experimental design. Data are drawn from 24 sessions of twelve traders each. We implement a 2×2 between-subjects design by varying the trading institution and the information precision of a signal. In the *Lit Only* market, subjects could trade only in the lit exchange, while in the *Dark* market, a dark pool is added alongside a lit exchange. For the low information precision environment, urn A contained six black balls and four white balls, while urn B contained four black balls and six white balls. For the high information precision environment, urn A contained seven black balls and three white balls, while urn B contained three black balls and seven white balls.

Treatment	Trading Institution	Information Precision	No. of Sessions	Total Subjects
<i>Lit Only-Low</i>	Lit Only	Low	6	72
<i>Dark-Low</i>	Dark	Low	6	72
<i>Lit Only-High</i>	Lit Only	High	6	72
<i>Dark-High</i>	Dark	High	6	72

displayed shares was updated on all traders' computer screens. During the trading period, traders could enter as many orders as they desire subject to the non-negative cash balance and short-selling constraints and cancel any of their unexecuted limit orders in the book at any time. All transactions were reported immediately to all traders, indicating the price and the transaction volume.¹³

Trades occurred whenever a trader entered a limit order that crossed with an existing limit order by stating a bid price greater than or equal to an existing ask or entering an ask price less than or equal to an existing bid. Partial executions of submitted limit orders were possible, and orders were executed following strict price and time priority rules. A share at an attractive price had priority over a share at a worse price.¹⁴ Within each price level, orders submitted at an earlier time were executed first.

Under the *Dark* trading institution, in addition to the MUDA market, which we refer to as the lit exchange, with the features discussed above, traders could submit their buy/sell offers to another market. In this second market, which we refer to as a dark pool, traders only submitted the shares of the asset that they wished to buy or sell.¹⁵ The active offers and transactions of a trader in the dark pool were visible only to that trader and no one else. Thus, unlike the lit exchange, where the order book was publicly displayed, and information on transactions was immediately updated, others' order submissions and transactions in the dark pool and the market depth were not revealed to traders.¹⁶

¹³Traders continuously observed on the screen their current position in terms of ECUs (cash) and shares of the asset, the number of shares they bought and sold, and the prices they paid for the shares they bought or sold. In addition, all past trading prices in the current period and the number of units transacted were continuously shown on subjects' screens.

¹⁴For example, a higher price for a buy order is more attractive. Similarly, a lower price for a sell order is more attractive.

¹⁵In the experiments, we used neutral terms for the markets. The MUDA market (or the lit exchange) was referred to as *Market X* and the dark pool was referred to as *Market Y*.

¹⁶Participants were told that their offers sent to the dark pool would be matched with another trader's offer confidentially and automatically by the computer whenever such a match exists. Partial

Traders who submit their orders to a dark pool can conceal their information and their trading intention to others and avoid having their orders directly impact the price in the lit exchange. One can imagine that a dark pool may appeal to an informed investor who wants to minimize the price impact due to information revelation. On the other hand, the drawback of submitting orders to a dark pool is that traders are not sure if a counterparty can match their orders, leading to execution risk. Even when a match exists and a transaction occurs, orders may not be executed at the best price.

Traders couldn't specify any price for the orders sent to the dark pool. Transaction prices in the dark venue were derived from the existing buy and sell offer prices in the lit exchange. Specifically, orders in the dark pool were executed at the (latest) mid-point of the best buy and sell offer prices in the lit exchange.¹⁷ This mid-point price was continuously updated on traders' screens so that they were aware of the potential price improvement offered by the addition of the dark venue.

To manipulate the precision of informative signals, we changed the composition of black and white balls in the urns. Specifically, for the low information precision environment, urn A contained six black balls and four white balls, while urn B contained four black balls and six white balls. For the high information precision environment, urn A contained seven black balls and three white balls, while urn B contained three black balls and seven white balls. In both low- and high information precision environments, drawing a black ball implies that it is more likely that the chosen urn is A. However, this likelihood is greater in a high information precision environment.

matches and executions were possible.

¹⁷Dark pools generally use three primary pricing rules: (1) mid-point of the best bid and offer, (2) derived pricing, for example, the average price during the last five minutes of trading in the lit exchange before the execution of the dark trade, and (3) negotiated pricing similar to the OTC trading where a pair of buyer and seller trade directly (Ye (2016)). In their study, Nimalendran and Ray (2014) find that while not all trades are at the mid-point of National Best Bid and Offer (NBBO), about 57% of transactions are within .01% of the price around the mid-point.

III. Results

A. Trading Volume and Liquidity

Is market trading volume and liquidity affected by the addition of a dark pool?

We start our analysis by studying the effect of dark trading on market trading activity and liquidity. Tables II and III report the results of an OLS regression of total transaction volume, the transaction volume in the lit exchange, and other liquidity measures for markets with low and high precision, respectively. The regressors include the variable *Dark* (which takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise), period, the sum of acquired signals, and the average values of the demographic variables in the market.¹⁸ The standard errors are clustered at the session level.

We find that the addition of a dark pool does not affect the aggregate transaction volume in the market under either a low or high-precision environment. This can be taken as a positive effect because this implies no alteration in market participation by investors with dark trading. However, the overall transaction volume in the lit exchange declines significantly in the presence of a dark pool. This is expected because, with market fragmentation, traders now use both lit and dark venues for their transactions. Given that the aggregate trade volume is similar with and without a dark pool, the dark pool now steals or substitutes transactions from the lit exchange.

¹⁸The demographic variables are *risk aversion* (a measure of how risk-averse a subject is; ranges from 1 to 11 corresponding to the respective subject's switching point in the Holt-Laury risk-elicitation procedure, with larger values indicating higher risk aversion), *age* (age of the participant in years), *male* (equals one if the participant is male and zero otherwise), and *Economics/Business major* (equals one if the subject is pursuing a business, or accountancy, or economics major).

Table II
OLS Regression of Total Transaction Volume and Liquidity Measures in the Lit Exchange in Low Precision Markets

This table presents results of OLS regression analysis of the total transaction volume in the market, transaction volume in the lit exchange, depth, quoted, and effective spread in a period in markets with low precision of signals. Depth is defined as the sum of all orders up to two points from the closing best bid, and best ask prices in a period. Quoted and effective spreads are defined as the (volume-weighted) average best bid-ask spread evaluated at each submission and transaction in a period, respectively. The baseline is the *Lit Only - Low* treatment. Standard errors (clustered at the level of independent session) are in parentheses. *Dark* takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise. All regressions include the sum of acquired signals in the market and average values of the demographic variables in a market as additional regressors. ** indicates significance at the 5% level, and *** indicates significance at the 1% level.

	Total Transaction Vol.	Transaction Vol. in Lit Exchange	Depth	Quoted Spread	Effective Spread
<i>Dark</i>	-8.44 (20.92)	-61.64*** (13.16)	-63.65** (24.33)	0.36 (0.39)	0.27 (0.34)
<i>Period</i>	-0.58 (0.52)	0.02 (0.79)	-0.25 (1.26)	0.01 (0.01)	-0.00 (0.01)
Constant	-102.70 (318.60)	117.50 (201.70)	-329.0 (659.6)	0.40 (7.11)	0.28 (5.99)
No. of obs.	240	240	240	239	238
No. of clusters	12	12	12	12	12
R^2	0.33	0.38	0.06	0.13	0.10

Table III
OLS Regression of Total Transaction Volume and Liquidity Measures in the Lit Exchange in High Precision Markets

This table presents results of OLS regression analysis of the total transaction volume in the market, transaction volume in the lit exchange, depth, quoted, and effective spread in a period in markets with high precision of signals. Depth is defined as the sum of all orders up to two points from the closing best bid, and best ask prices in a period. Quoted and effective spreads are defined as the (volume-weighted) average best bid-ask spread evaluated at each submission and transaction in a period, respectively. The baseline is the *Lit Only - High* treatment. Standard errors (clustered at the level of independent session) are in parentheses. *Dark* takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise. All regressions include the sum of acquired signals in the market and average values of the demographic variables in a market as additional regressors. * indicates significance at the 1% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level.

	Total Transaction Vol.	Transaction Vol. in Lit Exchange	Depth	Quoted Spread	Effective Spread
<i>Dark</i>	-11.98 (27.77)	-64.05** (27.00)	-69.67* (33.08)	0.68** (0.25)	0.47* (0.22)
<i>Period</i>	-0.63 (0.95)	0.26 (1.05)	1.09 (2.49)	0.01 (0.02)	-0.00 (0.01)
Constant	-671.5 (420.7)	-286.80 (439.1)	-756.00 (421.4)	12.19* (5.99)	13.03** (4.90)
No. of obs.	240	240	239	218	218
No. of clusters	12	12	12	12	12
R^2	0.11	0.20	0.10	0.17	0.15

This effect is observed in both low and high-precision markets.

RESULT 1: *When a dark pool is added alongside a lit exchange, compared to the lit-only market, the total transaction volume remains unchanged while there is a decline in the volume of transactions at the lit exchange. This observation holds irrespective of the precision of the information in the signals.*

Tables II-III also present OLS regression results using various liquidity measures. The following three measures of market liquidity are used: per-period depth, defined as the sum of all orders up to two points from the closing best bid and best ask prices in a period, quoted spread each period, defined as the (volume-weighted) average best bid-ask spread evaluated at each submission in a period, and effective spread each period, defined as the (volume-weighted) average best bid-ask spread evaluated at each transaction in a period.

The results show that, under a low precision environment, while there is an overall negative impact of dark trading on the depth at the exchange, both quoted and effective spreads remain largely unaffected. On the other hand, when the information precision in the market is high, the depth at the exchange goes down, and the spread widens, implying a significant adverse effect on overall liquidity at the exchange.

RESULT 2: *There is a negative impact of dark trading on liquidity at the lit exchange. The effects are more pronounced when the information precision of signals is high.*

B. Order Informativeness and Dark Pool Usage

How does the relative usage of the dark pool vis-à-vis the lit exchange depend on the relative strength of private information held by an investor?

We define an uninformed trader as one whose individual Bayesian posterior belief of state A equals 0.5. This includes the traders who cannot acquire or choose not to acquire information and, thus, by design, have an individual posterior equal to the prior and those traders who acquire information, but their information is not conclusive. Traders having an individual posterior different from 0.5 are referred to as informed traders. When there is only a lit market available for trading, the period-average informed-to-uninformed submission ratio equals 1.06 under low precision of signals and 1.24 under high precision environment. With the introduction of an additional dark venue for trading, this ratio equals 1.07 in the lit exchange and 1.43 in the dark pool when information precision is low. With high precision of signals, the informed-to-uninformed ratio becomes 1.30 in the lit exchange and 1.03 in the dark pool. This suggests that conditional on the information precision, the order informativeness at the lit exchange remains unchanged with the addition of a dark pool alongside a lit market.¹⁹

We classify the informed traders further into ones having either moderate signal (if the individual posterior lies in $[0.25, 0.50]$ or $(0.50, 0.75]$) or strong signal (if the individual posterior lies in $[0, 0.25]$ or $(0.75, 1]$). Table IV provides the composition of order submission volume in a market. Given that there are more traders with strong signals and fewer traders with moderate signals in a market with high information precision than the one with low precision of signals, the table shows that the corresponding entries for the moderately informed traders are smaller and for the strongly informed traders are greater under high precision.

More importantly, in markets with low precision, the composition of order sub-

¹⁹Order informativeness captures the relative informative content of submitted orders, and we use the informed-to-uninformed submission ratio at a trading venue as the measure of order informativeness. An alternative measure would be the fraction of total orders that are submitted by informed traders.

Table IV
Composition of Order Submissions

This table presents the percentage of submission volume in a market submitted by each type of trader- uninformed, moderately informed, or strongly informed. An uninformed trader has an individual Bayesian posterior belief of state A equal to the prior of 0.5. A moderately informed trader has an individual posterior that lies in $[0.25, 0.50)$ or $(0.50, 0.75]$, and a strongly informed trader has an individual posterior that lies in $[0, 0.25)$ or $(0.75, 1]$. The sum of the values in each column adds up to 100.

	<i>Lit Only</i> <i>-Low</i>	<i>Dark-Low</i>		<i>Lit Only</i> <i>- High</i>	<i>Dark-High</i>	
		Lit Exchange	Dark Pool		Lit Exchange	Dark Pool
Uninformed	48.3	48.2	43.2	44.1	44.0	46.8
Moderately Informed	27.9	26.5	30.8	18.4	12.6	14.4
Strongly Informed	23.8	25.3	26.1	37.5	43.4	38.8

missions at the lit exchange remains almost the same with only the lit market being present or when both venues for trading exist. On the other hand, when the precision of information is high, while the percentage of submitted orders at the lit exchange attributable to uninformed traders remains the same before and after the addition of a dark pool, the percentage of orders at the lit exchange that can be attributed to strongly (moderately) informed traders goes up (down) substantially. Therefore, although the informed-to-uninformed ratio does not vary at the lit exchange with or without a dark pool, a larger proportion of the informed orders are now coming from the investors with strong signals.

To better understand whether the strength of the individual posterior held by a trader influences her decision to use the dark pool relatively more, we conduct an OLS regression of two individual-level variables in the treatments with both trading venues: individual net dark submission volume defined as the submission volume in

the dark pool minus the submission volume at the lit exchange of a subject in a period, and individual dark pool usage defined as the ratio of dark submission volume over total submission volume for a subject in a period. The regressors include *Individual posterior strength*, defined as the absolute deviation of an individual trader’s posterior from 0.5, and the trading period. We also include several demographic variables as additional regressors.²⁰ The standard errors are clustered at the level of an individual subject. Table V reports the results of the regression analysis.

In the low precision environment, we observe no relationship between the strength of the individual posterior held by a trader and her relative dark pool usage. In contrast, when the information precision of signals is high, as the individual posterior deviates farther away from 0.5, compared to the benchmark of uninformed traders, an individual is more likely to use the lit exchange for order submissions. With high precision, many traders with strong signals are likely to cluster on the same side of the market, which raises the execution risk in the dark pool. Therefore, traders with greater posterior strength are more likely to utilize the lit exchange for faster execution. Consequently, the adverse selection risk for the uninformed traders becomes more elevated at the lit exchange than in the dark pool.²¹ Thus, we have the following result.

RESULT 3: *Traders with stronger signals have a relatively lower dark pool usage when the information precision is high.*

C. Prices: Information Aggregation, Forecasting, and Volatility

²⁰As defined earlier, these variables are *risk aversion*, *age*, *male*, and *Economics/Business major*.

²¹Given that informed traders utilize both lit exchange and dark pool, the risk of trading with someone with superior information is present in both venues. However, the exchange’s adverse selection risk is greater with the relatively higher usage of lit exchange by the traders with stronger signals.

Table V
**OLS Regression of Individual Net Dark Submission Volume
and Individual Dark Pool Usage: *Dark-Low* and *Dark-High*
Markets**

This table presents results of OLS regression analysis of the individual net dark submission volume in columns (1) and (3) and individual dark pool usage in columns (2) and (4). Individual net dark submission volume is defined as the submission volume in the dark pool minus the submission volume at the lit exchange of a subject in a period. Individual dark pool usage is defined as the ratio of dark submission volume over total submission volume for a subject in a period. Standard errors (clustered at an individual subject level) are in parentheses. *Individual posterior strength* is defined as the absolute deviation of an individual trader's posterior from 0.5. All regressions include the demographic variables as additional regressors. * indicates significance at the 10% level, and ** indicates significance at the 5% level.

	Low Precision		High Precision	
	(1)	(2)	(3)	(4)
<i>Individual posterior strength</i>	-42.34 (27.14)	-0.00 (0.10)	-47.33** (21.96)	-0.17** (0.08)
Period	-0.87** (0.43)	-0.00 (0.00)	-1.08** (0.43)	-0.00 (0.00)
Constant	-20.34 (40.95)	0.33* (0.18)	-38.77 (87.90)	0.24 (0.26)
No. of observations	1440	1387	1440	1402
No. of clusters	72	72	72	72
R^2	0.07	0.07	0.02	0.02

How does dark trading impact the ability of asset prices to reflect information available in the market and the propensity of prices to reveal the true state of nature?

To measure the fundamental value of the asset in our setting, for each market, we first calculate the Bayesian posterior probability of urn A given all draws in the market. This posterior multiplied by 10 provides the risk-neutral fundamental value of the asset. Note that this value differs across markets due to variations in the number of signals acquired and the difference in the information revealed by these signals. Taking the information-acquisition decisions as given, this Bayesian posterior times ten also gives the fully revealing rational expectations price. Figure 1 plots the expectation of the market price in the lit market conditional on the Bayesian posterior ($\mathbb{E}(\text{Price}|\text{Bayesian posterior})$) in each of the four treatments (see the long-dashed curve plotted in Figure 1). The risk-neutral fundamental value of the asset is depicted by the straight, short-dashed line starting from the origin.

Visual inspection of Figure 1 suggests that adding a dark pool does not worsen the ability of prices to aggregate diverse information present in the market. Average prices vary between a narrow range near the prior of 5, with and without a dark pool, when information precision of signals is low. On the other hand, a significant amount of available information is impounded into the transaction prices in the lit market in both treatments under the high information precision environment. For example, when the Bayesian posterior in the market is close to zero (ten), the average price is close to 2 (8) in both *Lit Only-High* and *Dark-High* treatments.

We calculate the following measures related to price quality (linear absolute deviation (LAD) and forecast error) and the volatility of prices.

- **LAD.** We define the linear absolute deviation (*LAD*) in a period as the absolute difference between the mean price in the lit market and the fully revealing price,

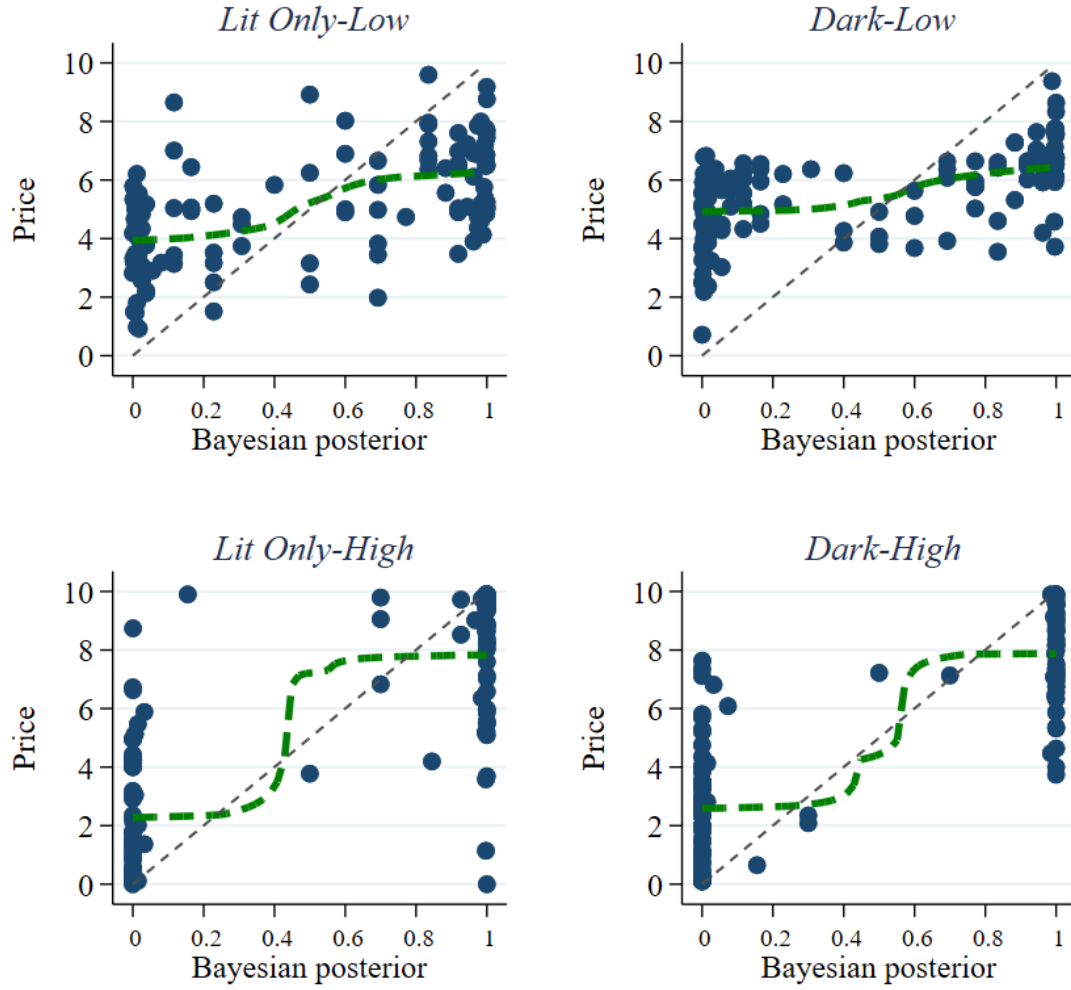


Figure 1. The precision of market prices in the lit market in each treatment. The long-dashed curve displays the average price in the lit market conditional on the Bayesian posterior in a market ($\mathbb{E}(\text{Price}|\text{Bayesian posterior})$). The estimation is computed by local linear regression using an Epanechnikov kernel bandwidth of 0.2. The risk-neutral fundamental value of the asset is depicted by the straight, short-dashed line starting from the origin.

that is, $LAD = |\text{Mean price} - 10(\text{Bayesian posterior})|$. The LAD measures price efficiency: how close transaction prices are to the fully revealing values.

- **Forecast error.** We define the forecast error for each period as the absolute difference between the mean transaction price and the asset’s true value. If the urn is A (B), the true value equals 10 (0). Forecast error measures how good the prices are in estimating the likelihood of the predicted event.
- **Volatility of prices.** We define the price volatility measure ($VOLA$) in a period by the standard deviation of log returns (Stöckl, Huber, and Kirchler (2015)):

$$\sqrt{\frac{1}{T} \sum_{t=1}^T (RET_t - \overline{RET})^2},$$

where $RET_t = \ln(P_t/P_{t-1})$, $\overline{RET}$ is the mean of log returns in a period, and T is the number of transactions in the lit exchange in a period.

To understand the effect of adding a dark pool alongside a lit exchange on the efficiency of asset prices and price volatility, we perform regression analyses with the above three measures as the dependent variable. The regressors include the variable *Dark*, total transactions in the lit market, and the average values of the demographic variables in the market. For the regression of forecast error, an additional regressor of *Bayesian posterior accuracy* is used that takes a value of 1 if the posterior in the market $> (<) 0.5$ and the underlying urn is A (B), and 0 otherwise. Table VI reports the results separately for the treatments with low and high precision of signals. The table shows that adding a dark pool increases the LAD and forecast error (and thus, is detrimental) when the information precision is low. At the same time, the availability of a dark pool lowers these measures (and thus, is beneficial) if the information precision is high. However, none of these coefficients are significant, and hence, we

conclude that, overall, the presence of a dark market does not affect the price quality measures. This is true irrespective of the precision of signals in the market.

RESULT 4: *Compared to the lit-only market, adding a dark pool does not affect price quality, irrespective of the information precision of signals.*

With the data subdivided by the level of dark trading in a market, Figures 2 and 3 plot $\mathbb{E}(\text{Price}|\text{Bayesian posterior})$ in the lit market of *Dark-Low* and *Dark-High* treatments, respectively. The corresponding plot for the benchmark *Lit Only-Low* treatment in Figure 2 and *Lit Only-High* treatment in Figure 3 are shown as well for ease of comparison. Defining the dark transaction ratio as the ratio of transaction volume in the dark pool over the total transaction volume in a period, the level of dark trading is sub-divided into the following three categories²²:

- Modest Dark Trading- when the dark transaction ratio is less than 0.2.
- High Dark Trading- when the dark transaction ratio exceeds 0.2 but is less than 0.4.
- Very High Dark Trading- when the dark transaction ratio exceeds 0.4.

When information precision is low, Figure 2 shows that average prices continue to vary within a narrow range near the prior of 5 for various levels of dark trading. Figure 3, on the other hand, indicates that compared to the *Lit Only-High* benchmark, prices are far more revealing with the addition of a dark pool when there is modest dark trading but not when the dark trading level is either high or very high.

Figure 4 further plots the average value of *LAD* in markets for each category (bin) of the level of dark trading ratio, separately for the markets with low and high preci-

²²Our observations in this section are robust if we use the 33rd and 66th percentiles instead in choosing the critical values to separate dark trading into modest, high, and very high.

Table VI
**OLS Regression of Price Quality Measures and Price
Volatility in the Lit Market**

This table presents the results of the OLS regression analysis of linear absolute deviation (*LAD*), forecast error, and price volatility (*VOLA*). The *LAD* is the absolute difference between the mean price in the lit market and the fully revealing price, that is, $LAD = |\text{Mean price} - 10(\text{Bayesian posterior})|$. Forecast error is given by the absolute difference between the mean transaction price and the asset's true value. *VOLA* is defined as the standard deviation of log returns. The baseline is the *Lit Only - Low* (*Lit Only - High*) treatment in Low (High) Precision markets. Standard errors (clustered at the level of independent session) are in parentheses. *Dark* takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise. *Bayesian posterior accuracy* takes a value of 1 if the posterior in the market $> (<)$ 0.5 and the underlying urn is A (B), and 0 otherwise. **All regressions include the total transactions in the lit market and the average values of demographic variables as additional regressors.** * indicates significance at the 10% level, ** indicates significance at the 5% level, and *** indicates significance at the 1% level.

	Low Precision			High Precision		
	<i>LAD</i>	Forecast Error	<i>VOLA</i>	<i>LAD</i>	Forecast Error	<i>VOLA</i>
<i>Dark</i>	0.34 (0.28)	0.21 (0.23)	-0.04 (0.06)	-0.23 (0.20)	-0.20 (0.19)	0.10* (0.05)
<i>Bayesian posterior accuracy</i>		-1.98*** (0.40)			-3.11*** (0.98)	
<i>Period</i>	-0.02 (0.01)	-0.02 (0.02)	0.00 (0.00)	-0.11*** (0.03)	-0.11*** (0.03)	0.01* (0.01)
Constant	-5.57 (5.03)	-1.09 (5.33)	0.22 (0.86)	-6.88** (2.76)	-2.85 (2.21)	1.51*** (0.46)
No. of observations	240	240	240	238	238	235
No. of clusters	12	12	12	12	12	12
R^2	0.06	0.18	0.09	0.28	0.32	0.07

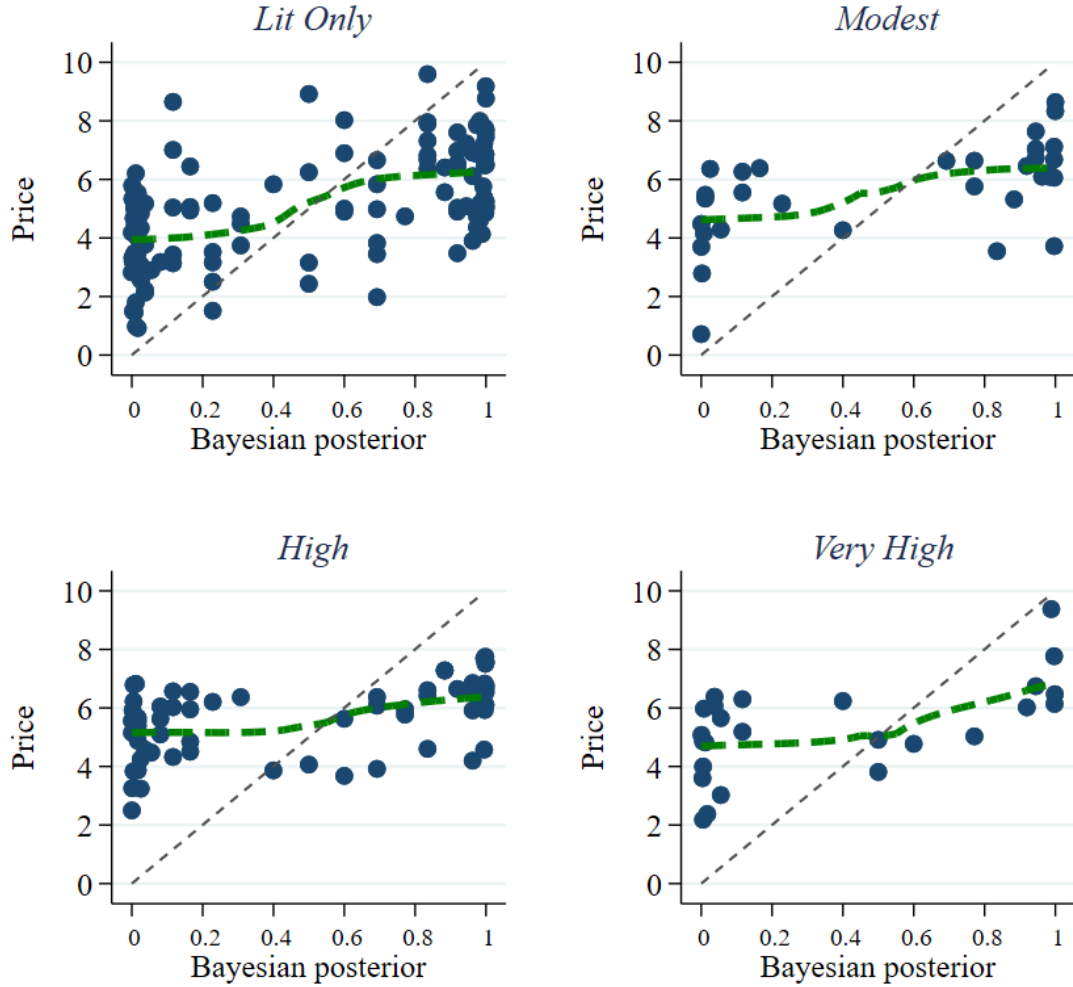


Figure 2. Precision of market prices in the lit market in *Lit Only-Low* and *Dark-Low* treatments. The data in the *Dark-Low* treatment is sub-divided by the level of dark trading: *modest*, *high*, and *very high*. The long-dashed curve displays the average price in the lit market conditional on the Bayesian posterior in a market ($E(\text{Price}|\text{Bayesian posterior})$). The estimation is computed by local linear regression, using Epanechnikov kernel bandwidth of 0.2. The risk-neutral fundamental value of the asset is depicted by the straight short-dashed line starting from the origin.

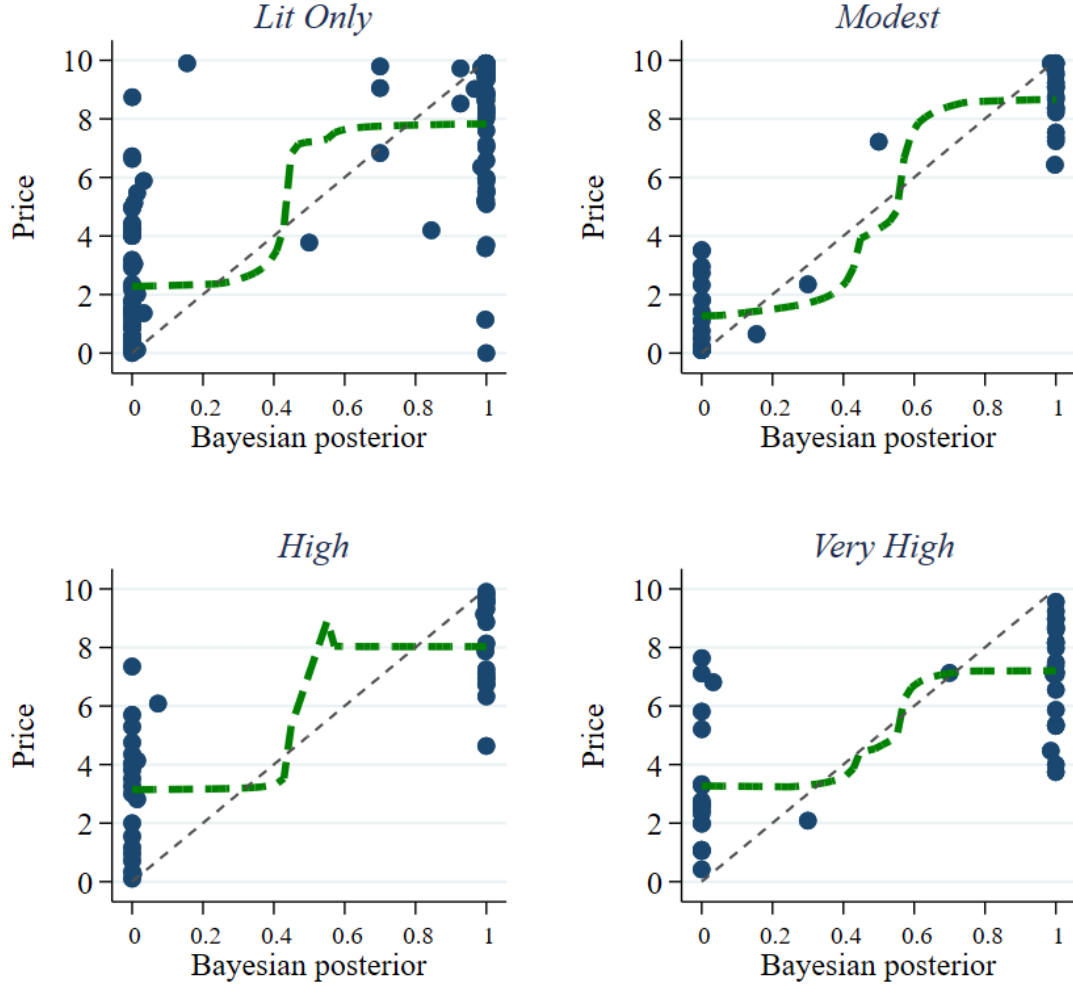


Figure 3. Precision of market prices in the lit market in *Lit Only-High* and *Dark-High* treatments. The data in the *Dark-High* treatment is sub-divided by the level of dark trading: *modest*, *high*, and *very high*. The long-dashed curve displays the average price in the lit market conditional on the Bayesian posterior in a market ($\mathbb{E}(\text{Price} | \text{Bayesian posterior})$). The estimation is computed by local linear regression, using Epanechnikov kernel bandwidth of 0.2. The risk-neutral fundamental value of the asset is depicted by the straight short-dashed line starting from the origin.

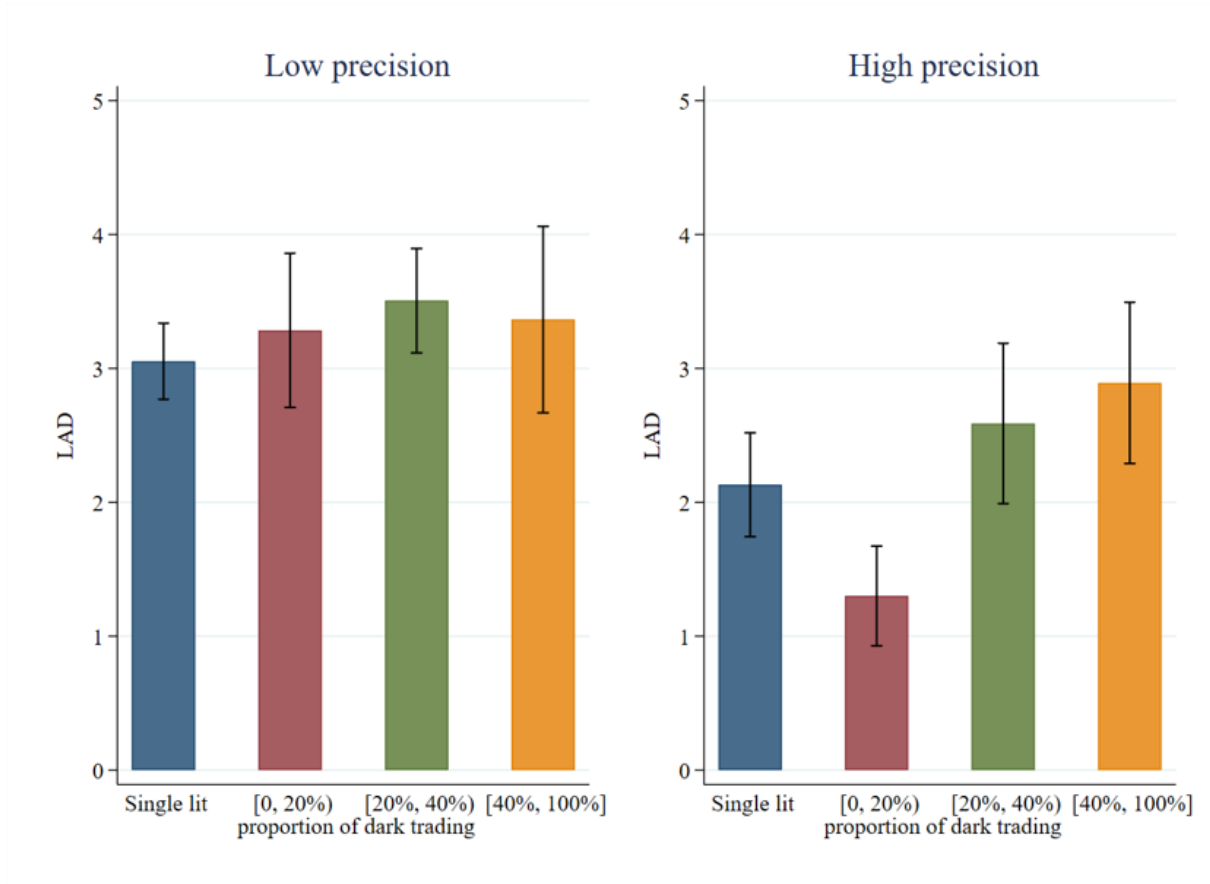


Figure 4. *LAD* subdivided by the level of dark trading. The *LAD* in markets in each category of the level of dark trading: *modest*, *high*, and *very high*. The left (right) panel is for markets with low (high) precision of signals. “Single lit” refers to the treatment with a single lit exchange only, i.e., *Lit Only-Low* and *Lit Only-High* in the left and right panels, respectively.

sion of signals. When information precision is low, there is no meaningful relationship between the *LAD* and the level of dark trading. However, when information precision is high, compared to the *Lit Only-High* benchmark, *LAD* is smaller with modest dark trading, similar or higher with high and very high dark trading. Therefore, we obtain a non-linear association between dark market participation and price efficiency.²³

RESULT 5: *When information precision of signals is high, compared to the lit-only*

²³A similar conclusion is reached if the forecast error is used instead of *LAD*.

market, the addition of a dark pool is associated with an improvement (deterioration) in price quality in the lit market when dark pool participation is modest (very high). There is a non-linear relationship between dark market participation and the ability of prices at the lit market to aggregate available information and reveal the underlying state of nature.

D. Information Acquisition

How does adding a dark pool affect the incentive to acquire costly information?

Table VII displays the results of regressing the number of signals acquired in the market on the variable *Dark*, the period number, and the average values of the demographic variables in the market. The standard errors are clustered at the session level. The regression results show that the presence of a dark venue encourages information acquisition when the information precision of signals in the market is high. However, in markets with low precision, there is no effect of adding a dark pool alongside a lit exchange; the coefficient is positive but insignificant. Therefore, we have the following result.

RESULT 6: *Adding a dark pool alongside a lit exchange increases costly information acquisition when the information precision is high.*

Table VII also reveals that, in general, information acquisition goes up as the experiment progresses in the low-precision environment. On the contrary, when the information precision is high, subjects acquire fewer signals over periods. Table VIII further displays the results of regressing the number of signals acquired in the market on the variable *High* (which takes a value of 1 if the treatment includes high precision

Table VII
OLS Regression of Number of Signals Acquired in a Market:
Effect of Dark Pool

This table presents the results of OLS regression analysis, with the dependent variable being the number of signals acquired in the market. The baseline is the *Lit Only - Low* treatment in the first column and the *Lit Only - High* treatment in the second column. Standard errors (clustered at the level of independent session) are in parentheses. *Dark* takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise. All regressions include the average values of the demographic variables in a market as additional regressors. * indicates significance at the 10% level, and ** indicates significance at the 5% level.

	Low Precision	High Precision
<i>Dark</i>	5.12 (4.72)	6.37** (2.64)
<i>Period</i>	0.31* (0.15)	-0.38* (0.20)
Constant	-59.96 (47.40)	-1.29 (29.32)
No. of observations	240	240
No. of clusters	12	12
R^2	0.33	0.32

Table VIII
OLS Regression of Number of Signals Acquired in a Market:
Effect of Information Precision

This table presents the results of OLS regression analysis, with the dependent variable being the number of signals acquired in the market. The baseline is the *Lit Only - Low* treatment in the first column and the *Dark-Low* treatment in the second column. Standard errors (clustered at the level of independent session) are in parentheses. *High* takes a value of 1 if the treatment comprises of markets with a high precision of signals and 0 otherwise. All regressions include the average values of the demographic variables in a market as additional regressors. *** indicates significance at the 1% level.

	Lit Only	Dark
<i>High</i>	-11.22*** (1.09)	-7.60*** (1.53)
<i>Period</i>	-0.27 (0.22)	0.20 (0.16)
Constant	-84.72*** (25.43)	73.05*** (16.74)
No. of observations	240	240
No. of clusters	12	12
R^2	0.51	0.58

of signals and 0 otherwise), the period number, and the average values of the demographic variables in the market. The standard errors are clustered at the session level. The table shows that a higher precision of signals discourages information acquisition irrespective of the presence of a dark pool. Conditional on the market trading institution, the acquired signals are relatively more informative when the signal precision is high. Therefore, traders would need to acquire fewer signals now when compared to the low-precision environment.

RESULT 7: *Conditional on the market trading institution, higher information precision of signals results in the acquisition of fewer signals.*

E. Trader's Earnings

Are informed traders better off with the introduction of a dark pool? Are earnings higher with a relatively higher dark pool usage?

We calculate the (gross) profits of trader i in period t as $\Delta ECU_{it} + 10\Delta Assets_{it}$ if the urn was A and ΔECU_{it} if the urn was B, where ΔECU_{it} measures the difference between cash endowment at the end (post-trade) and the start (pre-trade) of a period before subtracting the information-acquisition costs, and $\Delta Assets_{it}$ denotes the stock balance at the end of a period minus the initial stock endowment. Thus, gross profit is the difference between the value of a trader's portfolio at the end and at the start of a period (without subtracting the cost of information acquisition).

With gross profits as the dependent variable, we perform OLS regression analysis to understand whether informed traders outperform uninformed ones, whether strongly informed traders earn more, and whether adding a dark pool increases the overall earnings of any type of trader. Various specifications are considered, and the results are reported in Table IX. Columns (1) and (3) show that informed traders are able to outperform the uninformed ones in markets with low precision as well as high precision. In general, the extent of outperformance is greater when information precision is low than when it is high.²⁴ Columns (2) and (4) further indicate that traders with strong signals can reap substantial benefits when compared to the uninformed ones. However, moderately informed traders get profits similar to those of uninformed traders.²⁵

²⁴As the table shows when there is only a lit market, informed traders' gross profits are higher by 47.79 points in *Lit Only-Low* and by 28.68 points in *Lit Only-High* treatment. Further calculations show that the gross profits of the informed traders exceed by 54.34 (41.75) points in *Dark-Low* (*Dark-High*) treatment, both values being significant at the 1% level.

²⁵The earnings of the moderately informed traders exceed the uninformed ones by 18.62 (significant at only 10% level) under *Dark-Low* and fall short by 1.28 (but insignificant) under *Dark-High*

Table IX effectively provides evidence for two critical observations. First, the availability of an additional dark venue for trading in itself does not affect the earnings of each type of trader, and those with stronger signals are able to outperform uninformed ones comfortably irrespective of the trading institution. Second, uninformed traders have a significant adverse selection risk while trading with investors with strong signals. This is present in all markets, with and without the addition of a dark pool, irrespective of low or high precision of signals.²⁶

To understand how the relative transactions of an individual trader in the dark market affect her gross earnings, we conduct an OLS regression of gross profits for each type of trader- uninformed, moderately informed, and strongly informed, using data from the *Dark-Low* and *Dark-High* treatments. Table X reports the results. While informed traders' earnings are not affected significantly by their relative use of the dark pool vis-à-vis the lit exchange, uninformed traders with a higher net dark transaction volume obtain higher profits. Therefore, lower profits are associated with uninformed traders having a higher immediacy demand. Although not readily apparent, one of the reasons could be that uninformed traders face a higher adverse selection risk at the lit exchange than at the dark pool. The addition of a dark pool allows them to use the additional trading venue to mitigate part of the adverse selection risk.

RESULT 8: *Uninformed traders who trade relatively more in the dark pool have higher earnings.*

treatment.

²⁶We note that although the difference in the profits of the strongly informed and uninformed is more elevated in markets with low precision, there are fewer such traders with strong signals compared to markets with high precision. There is greater competition among traders with strong signals in the latter markets.

Table IX
OLS Regression of Individual Gross Profits

This table presents the results of OLS regression analysis of the individual gross profits, defined as the difference between the values of trader portfolios at the end and the beginning of each period, excluding the cost of information acquisition. Standard errors (clustered at an individual subject level) are in parentheses. The baseline is the individual gross profits obtained by uninformed traders in *Lit Only-Low* treatment in columns (1) and (2) and in *Lit Only-High* treatment in columns (3) and (4). *Informed* takes a value of one if the trader is informed, i.e., has an individual Bayesian posterior different from 0.5. *Moderately Informed* takes a value of one if the trader is moderately informed, i.e., the individual posterior of the trader lies in $[0.25, 0.50]$ or $(0.50, 0.75]$. *Strongly Informed* takes a value of one if the trader is strongly informed, i.e., the individual posterior of the trader lies in $[0, 0.25]$ or $(0.75, 1]$. *Dark* takes a value of 1 if the treatment includes a dark pool along with the lit exchange and 0 otherwise. All regressions include the demographic variables as additional regressors. *** indicates significance at the 1% level.

	Low Precision		High Precision	
	(1)	(2)	(3)	(4)
<i>Informed</i>	47.79*** (12.92)		28.68*** (7.66)	
<i>Moderately Informed</i>		11.15 (12.45)		4.72 (11.67)
<i>Strongly Informed</i>		100.10*** (19.35)		41.28*** (8.43)
<i>Dark</i>	-3.55 (9.37)	-3.44 (9.39)	-5.13 (5.81)	-5.19 (5.80)
<i>Informed</i> $\times$ <i>Dark</i>	6.56 (16.35)		13.07 (11.15)	
<i>Moderately Informed</i> $\times$ <i>Dark</i>		7.47 (16.63)		-6.00 (16.52)
<i>Strongly Informed</i> $\times$ <i>Dark</i>		-0.02 (23.44)		15.00 (12.68)
Constant	-15.45 (41.60)	-10.74 (39.81)	31.46 (32.42)	31.07 (30.90)
No. of observations	2880	2880	2880	2880
No. of clusters	144	144	144	144
R^2	0.02	0.05	0.02	0.03

Table X
OLS Regression of Individual Gross Profits: *Dark-Low* and *Dark-High* Markets

This table presents the results of the OLS regression analysis of the individual gross profits in a period. Standard errors (clustered at an individual subject level) are in parentheses. *Individual net dark transaction volume* is defined as the transaction volume in the dark pool minus the transaction volume at the lit exchange of a subject in a period. All regressions include the demographic variables as additional regressors. * indicates significance at the 10% level, and ** indicates significance at the 5% level.

	Low Precision			High Precision		
	Uninformed	Moderately Informed	Strongly Informed	Uninformed	Moderately Informed	Strongly Informed
<i>Individual net dark transaction volume</i>	0.71** (0.35)	0.03 (0.46)	0.42 (0.36)	0.46** (0.22)	0.45 (0.49)	-1.02 (1.30)
Constant	-44.50 (79.69)	-176.70** (80.44)	223.90* (124.20)	65.24 (52.68)	-135.4 (136.60)	123.4 (116.10)
No. of observations	639	369	289	630	158	450
No. of clusters	66	47	44	66	31	45
R^2	0.02	0.01	0.02	0.03	0.02	0.06

IV. Discussion

The theoretical prediction regarding the self-selection of venues and sorting of traders based on their private information finds support in our data when the information precision of signals is high. Although informed traders in our experiment do not have perfect information, when the information precision of signals is high, the strongly informed traders have close to perfect information. These traders find themselves on the same side of the market and, consequently, suffer lower execution probabilities in the dark pool. As a result, they are more likely to migrate to the lit exchange for faster execution.

The relationship between the self-selection of venues by traders and price discovery is not as apparent as suggested by theory. As shown in Table VI, on average, adding a dark pool neither promotes nor impairs price discovery on the exchange.²⁷ Though there is no overall effect of the addition of a dark pool on price discovery, traders with stronger signals find it extremely difficult to execute their orders in the dark pool and populate the lit exchange relatively more when the dark market participation is modest in the market with high information precision, thereby impounding information into asset prices. Therefore, we find support for the theoretical hypothesis that adding a dark pool improves price discovery when the information precision is high, but only if the dark market participation is below a threshold.

The self-selection result of the theoretical models implies that adding a dark pool would discourage information acquisition when the information precision of signals is sufficiently high. However, we find the opposite: dark trading encourages costly information acquisition among investors. We believe there are two reasons for this observation. First, investors would anticipate that a sizeable number of orders will

²⁷This observation is similar to the finding of Bloomfield, O'Hara, and Saar (2015), who report that in a single-limit order book environment, the ability to hide orders affects traders' strategies but does not affect informational efficiency.

now be traded off-exchange given that markets are fragmented, resulting in less information being impounded into prices of transactions at the exchange. This would incentivize traders to acquire a larger number of signals. This is because, as shown in Table IX, information is precious irrespective of whether precision is low or high, with informed traders outperforming the uninformed ones. Given that prices are not fully revealing in markets with low precision, and even when they fully reveal the available information in several of the markets with high information precision, it takes time to do so, informed traders are able to outperform uninformed ones, and incentive to acquire information is present in all our markets (Grossman (1976), Grossman and Stiglitz (1980), Verrecchia (1982)).

Second, it seems likely that some traders would acquire more information in the hope that they will trade in the dark pool and be able to prevent the information leakage that would have resulted from placing orders in the lit exchange. Whether or not these traders are able to use the dark pool for performing their transactions though depends on several endogenous factors, like the execution risk and wait time in the dark pool. It is likely that if orders have to wait much longer for execution in the dark pool, then the ongoing mid-point price may change adversely by the time the order is finally matched to a counterparty offer, if at all it is executed.

The predictions of the theory, however, do not hold when the information precision of signals is low. In this case, the proportion of traders with strong signals in the market is much lower, and information risk is substantial, and hence the *sorting effect* disappears.

In summary, our experiments clearly show that the impact of trading in dark pools on information acquisition, informational efficiency of prices, and market outcomes depends crucially on the informativeness of the signals acquired by investors. As listed in results 1-2, there is a strong negative *crowding out effect* on liquidity

at the lit exchange irrespective of the information precision of signals. At the same time, there are two positive effects of the addition of a dark pool: *information acquisition effect* whereby investors acquire a greater number of signals, and a *sorting effect* wherein traders with stronger signals use the lit exchange relatively more than either traders with moderate signals or uninformed ones. These positive effects are significant only when the information precision is sufficiently high (results 3 and 6). With low precision of signals, the *information acquisition effect* and the *sorting effect* are non-existent or mild at best.

With high precision of signals, at a modest level of dark trading, the combination of *information acquisition effect* and *sorting effect* outweighs the *crowding out effect*. When the eventual dark market participation in the market is at a high level, the *crowding out effect* is stronger and exceeds the positive effects. Consequently, a non-linear relationship between the informational efficiency of prices and the level of dark trading is observed.²⁸ In contrast, when information precision is low, none of the positive effects are observed. Even then, price efficiency does not decline significantly with dark trading and deteriorates when dark market participation is very high.

Finally, in the context of laboratory markets, most studies use the centralized double auction trading institution with an open order book (Sunder (1995), Deck and Porter (2013), Noussair and Tucker (2013), Palan (2013)). Does an increase in opacity necessarily lead to the deterioration of prices as information aggregation tools? Recent experimental studies suggest that the absence of pre-trade transparency may not be detrimental for information aggregation in markets with hidden liquidity (Bloomfield, O'Hara, and Saar (2015), Gozluklu (2016)) as well as in decentralized markets (Asparouhova and Bossaerts (2017), Asparouhova, Bossaerts, and Yang (2024)). Our research demonstrates that when the information precision of acquired signals is suf-

²⁸Overall, the addition of a dark pool does not cause the price quality to deteriorate, as can be seen from Table VI.

ficiently high, introducing dark pools opens up additional channels, viz., the *information acquisition effect* and *sorting effect* that positively influence the informational efficiency of prices, although liquidity at the exchange is impacted negatively.

V. Conclusion

In modern financial markets, investors can trade not only in traditional lit exchanges but also in alternative trading venues such as dark pools. We investigate the effects of adding a dark pool alongside a lit exchange on the incentives to acquire costly information, price efficiency, and market liquidity. We report data from a series of laboratory markets for an asset whose terminal payoff is contingent on an unknown state of the world. Prior to trading, investors can purchase noisy signals at a cost. In addition to a lit exchange organized as a multiple-unit double auction market, traders can send their orders to a dark pool where others do not publicly observe their offers. These offers are executed at prices derived from the lit exchange.

The effect of dark trading on the efficiency of asset prices and market quality depends on the information precision of the acquired signals. When informative signals have high precision, the presence of a dark venue promotes information acquisition. This, in turn, positively influences the price discovery process in a financial market if the dark market participation is below a certain threshold and harms price efficiency only for a very high level of dark trading. While evaluating the pros and cons of dark trading, our study strongly indicates that regulatory authorities must consider the effect of dark pools on information acquisition. This bright side of dark markets should not be overlooked.²⁹

²⁹Although trading is a zero-sum activity in our setup, information acquisition can be regarded as a beneficial act, as without any information acquisition, prices cannot reveal the underlying state of nature. Therefore, the forecasting ability of prices is likely to be better with higher information acquisition even though the acquisition of such information is costly for an individual trader.

Given our research questions and as a starting point for studying the impact of introducing a dark pool running parallel to a limit order market in experimental asset markets, we assume that all investors are identical in their initial endowments of cash and shares of assets. It would be worthwhile to explore the implications of allowing heterogeneous endowments to capture the notion of institutional investors who can potentially transact using the dark venue without impacting the lit market with their large orders. Relatedly, other variations of our design include changing the price of the matched orders in the dark pool, varying the tick size, introducing exogenous caps on dark trading, and implementing different execution priority rules for orders sent to dark pool.³⁰

³⁰Instead of the orders in the dark pool being matched at the mid-point of the best buy and best sell offer prices in the lit exchange, dark orders can be designated as passive or aggressive. A passive buy (sell) order can be matched at the best buy (sell) offer price, while an aggressive buy (sell) order can be matched at the best sell (buy) offer price.

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